

Summary of Notable Supplements



Supplement	Primary Effects	Strength of Evidence	Dose	Population Most Likely to Benefit
Creatine monohydrate	Strength, power, lean mass	Strong	0.1 g/kg/day	Anyone interested in increasing strength, power, lean mass
Caffeine	Power and aerobic performance	Strong	3–6 mg/kg	Tired exercisers or competitive athletes seeking marginal improvements in power or aerobic performance
Carbohydrate	Moderate-to-high volume glycolytic performance	Strong	Scaled to exercise volume; often between 5–12 g/kg/day	Anyone engaged in high-volume or high-intensity exercise
Protein	Lean mass	Strong	1.6–2.2 g/kg/day	Anyone interested in increasing strength, power, lean mass
Essential amino acids	Lean mass	Strong	6–15 g	Anyone interested in increasing strength, power, lean mass with inadequate total protein intake
Sodium bicarbonate	Glycolytic performance	Strong	0.2–0.4 g/kg	Competitive athletes seeking marginal benefits in sports emphasizing short- to intermediate-distance sprinting
Beta-alanine	Glycolytic performance	Strong	4–6 g/day	Competitive athletes seeking marginal benefits in sports emphasizing short- to intermediate-distance sprinting
Dietary nitrate	Power, strength-endurance, and aerobic performance	Strong	400–800 mg	Competitive team sport athletes and endurance athletes
Branched chain amino acids	Soreness, recovery	Moderate/Mixed	10–15 g/day	Athletes with impaired recovery or actively engaged in unaccustomed exercise inducing considerable muscle damage

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HMB	Lean mass, recovery	Moderate/Mixed	3 g/day	Athletes with impaired recovery or actively engaged in unaccustomed exercise inducing considerable muscle damage
Phosphatidic acid	Lean mass	Moderate/Mixed	750 mg/day	Competitive athletes seeking marginal increases in strength or hypertrophy
Citrulline	Strength-endurance and aerobic performance	Moderate/Mixed	4–6 g	Competitive athletes seeking marginal improvements in strength-endurance
Glycerol	Aerobic performance	Moderate/Mixed	1.2 g/kg	Competitive athletes seeking improved hydration status or marginal improvements in aerobic performance
Capsaicin	Strength-endurance, aerobic performance	Moderate/Mixed	12 mg	Competitive athletes seeking marginal improvements in strength-endurance
Quercetin	Aerobic performance, recovery	Moderate/Mixed	1 g/day	Competitive athletes seeking marginal improvements in aerobic performance
Taurine	Aerobic performance, recovery	Moderate/Mixed	1–3 g/day	Competitive athletes seeking marginal improvements in aerobic performance
Arginine	Strength-endurance, aerobic performance	Weak/Lack of Evidence	n/a	n/a
ATP	Strength-endurance, exercise capacity	Weak/Lack of Evidence	n/a	n/a
Alpha-ketoglutarate	Strength-endurance, aerobic performance	Weak/Lack of Evidence	n/a	n/a

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Prohormones	Lean mass, strength	Weak/Lack of Evidence	n/a	n/a
Carnitine	Fat oxidation, aerobic performance, recovery	Weak/Lack of Evidence	n/a	n/a
Deer Antler Velvet	Lean mass, recovery	Weak/Lack of Evidence	n/a	n/a
Glutamine	Lean mass, recovery	Weak/Lack of Evidence	n/a	n/a
MCTs	Fat oxidation, aerobic performance, recovery	Weak/Lack of Evidence	n/a	n/a
Ribose	Exercise capacity	Weak/Lack of Evidence	n/a	n/a
Fat burners	Fat loss	Weak/Lack of Evidence	n/a	n/a
Androstenedione	Lean mass, strength	Banned	n/a	n/a
Dimethylamylamine (DMAA)	Stimulant (alertness, acute performance)	Banned	n/a	n/a
1,3-dimethylbutylamine (DMBA)	Stimulant (alertness, acute performance)	Banned	n/a	n/a
Note: All dosages listed as absolute dose (in grams or milligrams) or relative dose scaled to total body mass (g/kg or mg/kg).				